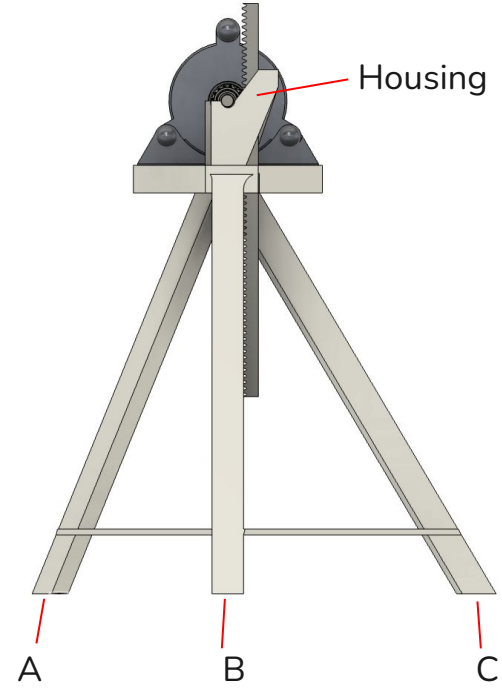
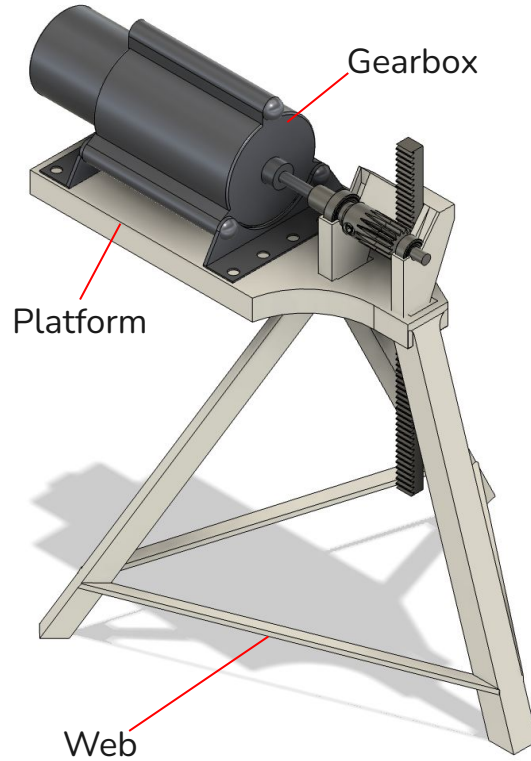
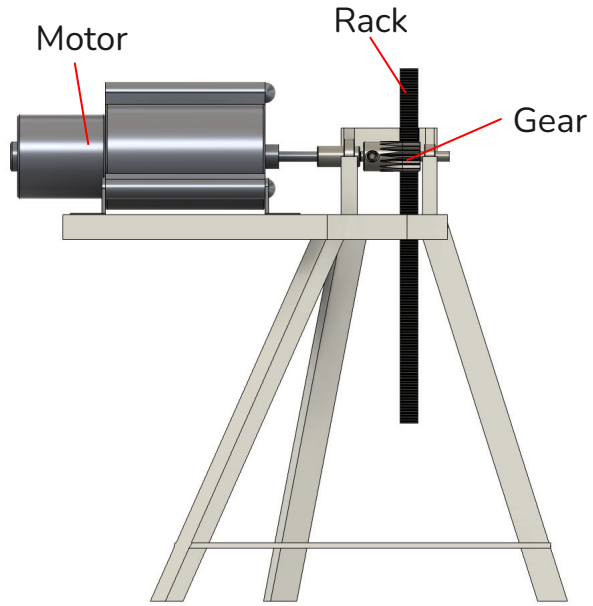
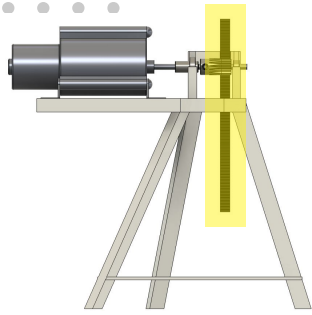


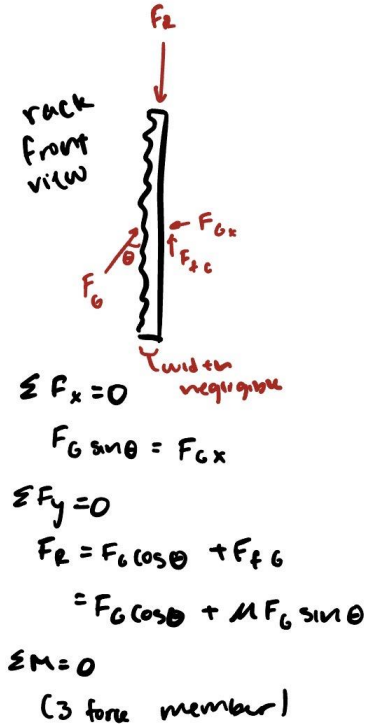
Design



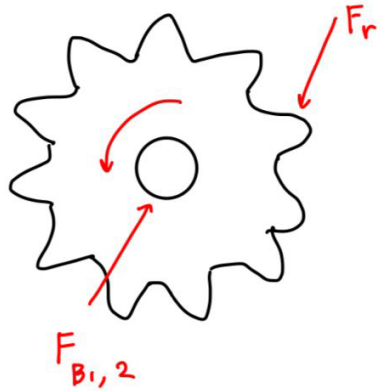


Rack FBD

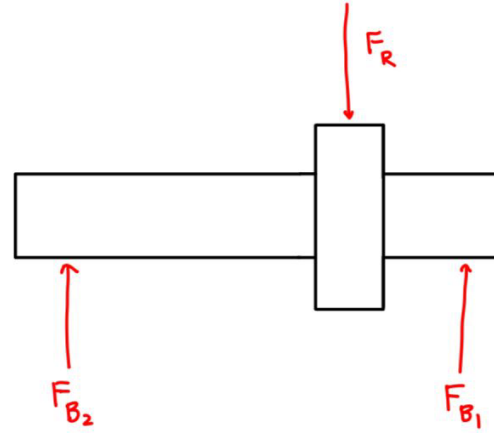
- 3 force member, needs horizontal support
- F_{GX} must act higher than F_G !



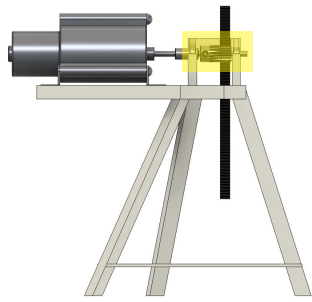
Transmission FBD



$$\tau_m = F_r r_{\text{gear}}$$
$$F_{B1} + F_{B2} = F_r$$

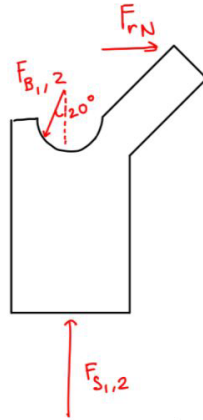


Shaft in bending



Housing FBD

Front view



Pressure angle = 20°

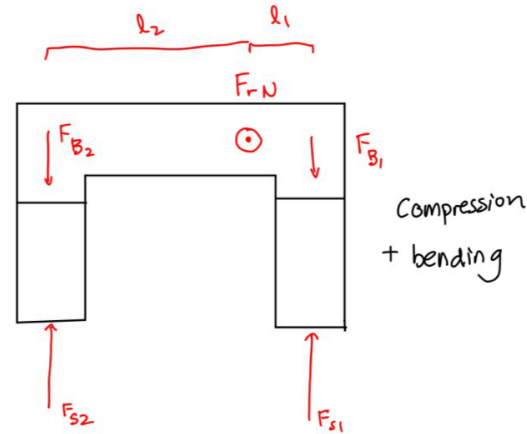
$$\sum F_x = -F_{B1,2} \sin(20^\circ) + F_{RN} = 0$$

$$F_{B1,2} \sin(20^\circ) = F_{RN}$$

$$\sum F_y = F_{S1,2} - F_{B1,2} \cos(20^\circ) = 0$$

$$F_{S1,2} = F_{B1,2} \cos(20^\circ)$$

side view



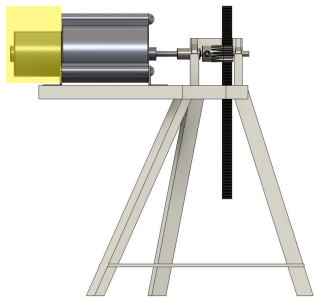
Compression
+ bending

$$\sum F_x = 0$$

$$\sum F_y = -F_{B2} - F_{B1} + F_{S2} + F_{S1}$$

$$\sum M = F_{B1} \cdot l_1 - F_{S1} \cdot l_1 - F_{B2} \cdot l_2 + F_{S2} \cdot l_2 = 0$$

$$F_{S1} \cdot l_1 + F_{B2} \cdot l_2 = F_{B1} \cdot l_1 + F_{S2} \cdot l_2$$



Motor FBD

motor
front
view:



$$\sum F_x = 0$$

$$\sum F_y = 0$$

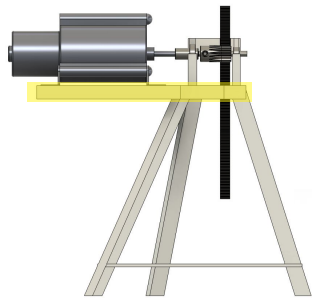
$$F_{T1} = F_{T2} (= F_T)$$

$$\sum M = 0$$

$$T = F_T (2r)$$

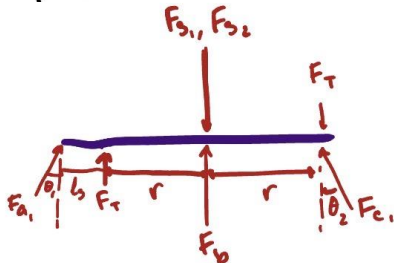
- Needs force couple to react shaft torque
- 34 N couple at stall—significant effect!





Platform FBD

platform front view



$$\sum F_x = 0$$

$$F_{a1} \sin \theta_1 = F_{c1} \sin \theta_2$$

$$\sum F_y = 0$$

$$F_{s1} + F_{s2} = F_b + F_{a1} \cos \theta_1 + F_{c1} \cos \theta_2$$

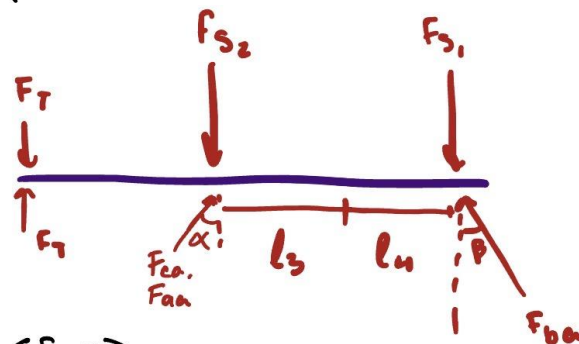
$$\sum M = 0$$

$$@ F_b \quad F_{c1} \cos \theta_2 r + 2F_T r = F_{a1} \cos \theta_1 (l_3 + r)$$

$$@ F_c \quad F_b r + 2F_T r + F_{a1} \cos \theta_1 (l_3 + r) = (F_{s1} + F_{s2}) r$$

$$@ F_a \quad F_T l_3 + F_b (r + l_3) + F_{c1} \cos \theta_2 (2r + l_3) = F_T (2r + l_3) + (F_{s1} + F_{s2}) (l_3 + r)$$

platform side view



$$\sum F_x = 0$$

$$(F_{ca} + F_{ba}) \sin \alpha = F_{ba} \sin \beta$$

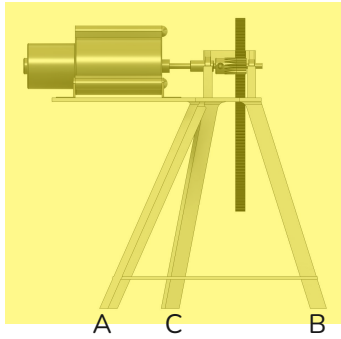
$$\sum F_y = 0$$

$$F_{s2} + F_{s1} = (F_{ca} + F_{ba}) \cos \alpha + F_{ba} \cos \beta$$

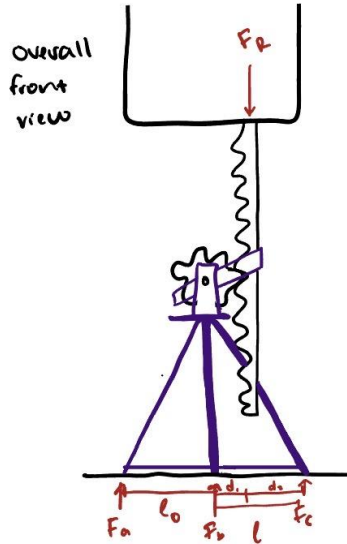
$$\sum M = 0$$

$$@ F_{s2} \quad F_{ba} \cos \beta (l_3 + l_4) = F_{s1} (l_3 + l_4)$$

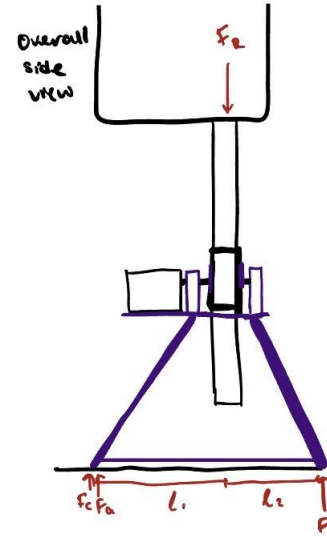
$$@ F_{s1} \quad (F_{ca} + F_{ba}) \cos \alpha (l_3 + l_4) = F_{s2} (l_3 + l_4)$$



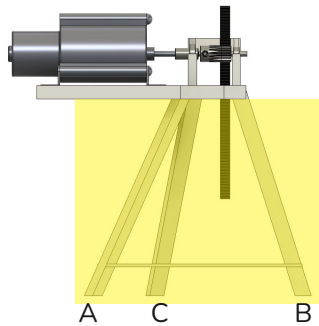
Overall FBD



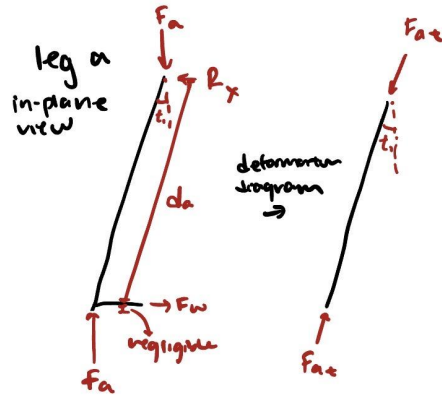
$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ F_R &= F_A + F_B + F_C \\ \sum M &= 0 \\ @ F_A \quad F_R(d_1 + l) &= F_B l + 2F_C l \\ @ F_B \quad F_R d_1 + F_A l_0 &= F_C l \\ @ F_A \quad F_A(l_0 + d_1) + F_B d_1 &= F_C d_2 \\ @ F_C \quad 2F_A(l_0 + l) + F_B l &= F_A d_2\end{aligned}$$



$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ F_R &= F_C + F_A + F_B \\ \sum M &= 0 \\ @ F_C / F_A \quad F_R l_1 &= F_B(l_1 + l_2) \\ @ F_B \quad F_R l_2 &= (F_C + F_A)(l_1 + l_2) \\ @ F_R \quad (F_C + F_A) l_1 &= F_B l_2\end{aligned}$$



Leg and Web FBD



$$\sum F_x = 0$$

$$F_{ax} \sin t_1 = F_{cx} \sin t_1$$

$$\sum F_y = 0$$

$$F_{ay} \cos t_1 = F_{cy} \cos t_1$$

$$\sum M = 0$$

(2 force member)

$$F_{wa} = F_{wb}$$

$$\sum F_x = 0$$

$$F_{wa} = F_{wb} \quad ** F_{wa} \approx F_{wb} \approx F_{wc}$$

$$\sum F_y = 0$$

$$\sum M = 0$$

- Qualitatively same for each leg
- Loads are asymmetric

Motor Characteristics

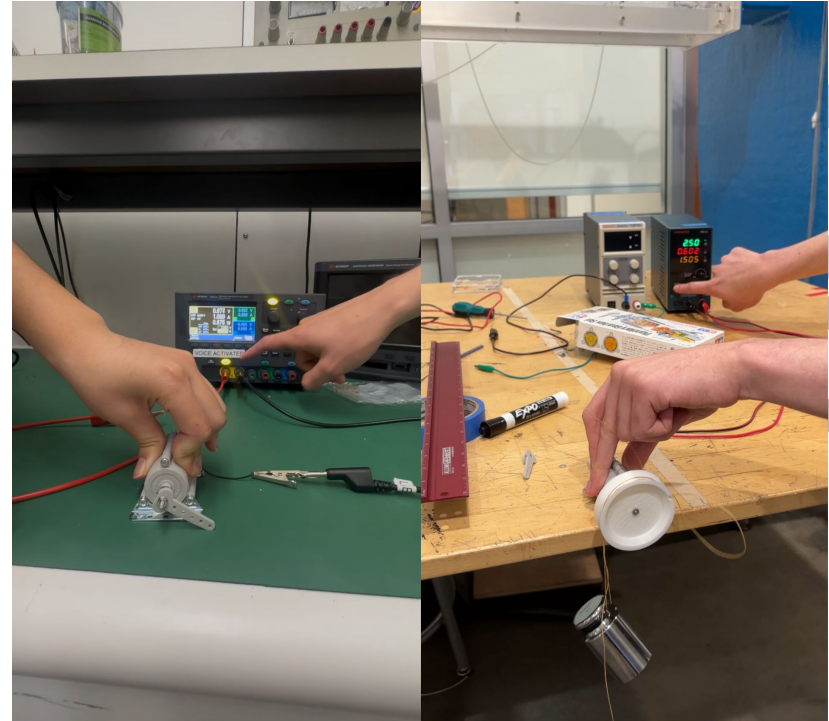
2.5V applied for all tests.

Motor constant (k): 0.0022
V/rad/s

Coil resistance (R): 0.88 Ω

Friction torque: 0.00055 N·m

Gearbox efficiency: 0.6



Motor Set Point

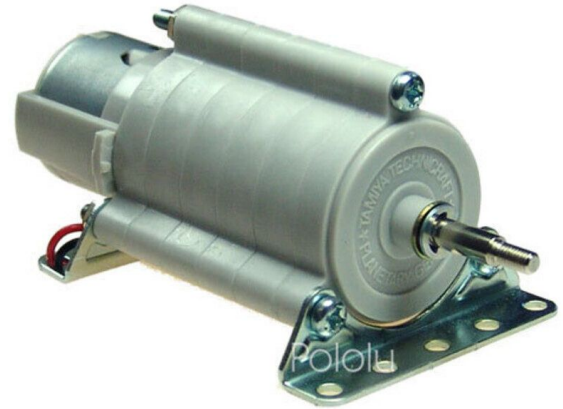
$$I_{\text{stall}} = 2.83 \text{ A (thermal max is 9.5)}$$

$$T_{\text{stall}} = 1.36 \text{ Nm}$$

$$\frac{1}{2} I_{\text{stall}} \rightarrow P_{\text{peak}} \rightarrow T_{@2.5 \text{ V}} = 0.62 \text{ N} \cdot \text{m}$$

$$\dot{\theta} = 1.41 \text{ rad/s}$$

Gear Ratio: 400:1 `



Gear & Rack Properties



$V_{\text{rack}} = 0.5 \text{ cm/s}$ yields:

$$r_{\text{gear}} = 0.36 \text{ cm}$$

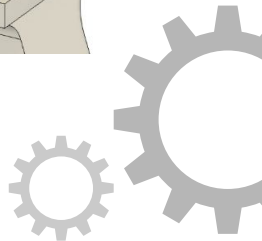
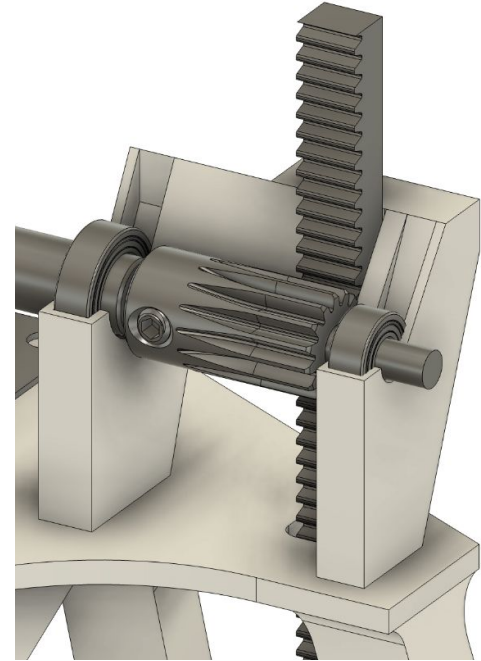
$$R_{\text{transmission}} = 0.36 \text{ rad/cm}$$

Rack & Pinion Selection

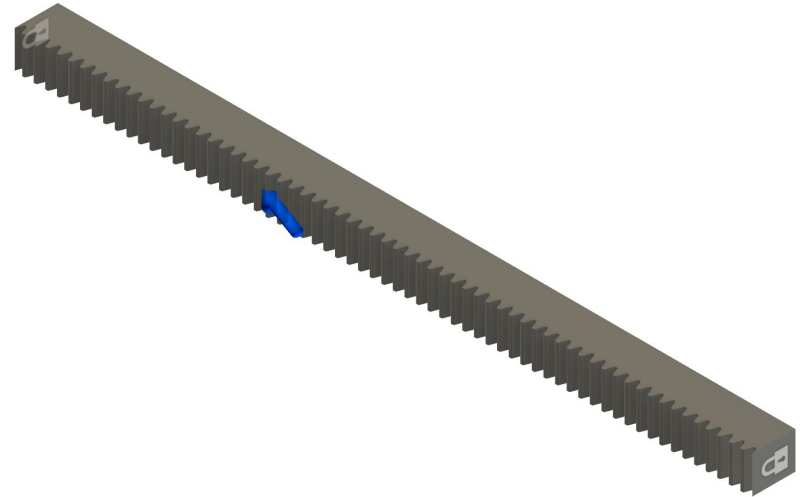
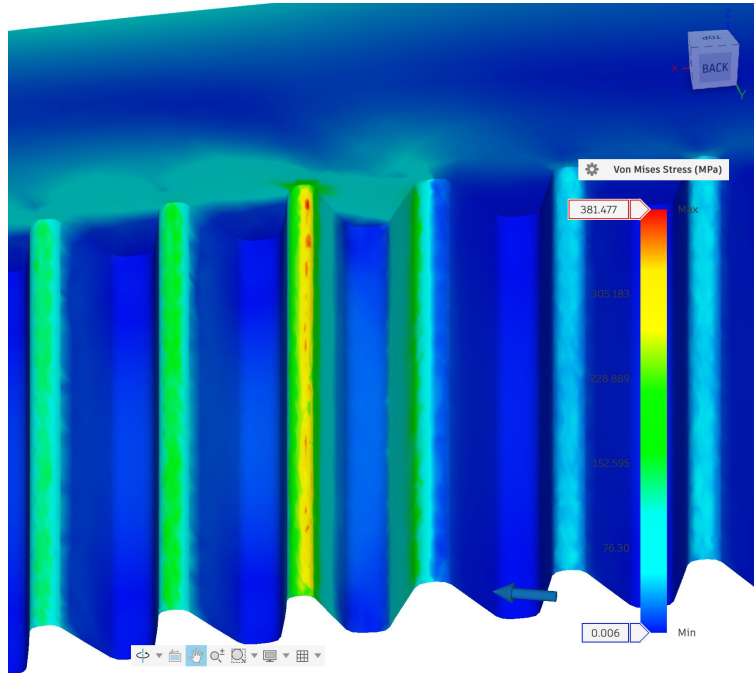
Rigid 10K $\rightarrow w_{\text{gear}} = 12 \text{ cm} \rightarrow \text{Impractical}$

Stock Rack & Pinion (treated carbon steel)

- $w_{\text{rack}} = 5 \text{ mm}$
- $r_{\text{gear, pitch}} = 3.75 \text{ mm}$
- $FS_{\text{pinion, stall}} = 1.6$



Rack & Pinion Confirmation



Material Selection (Structure)

$$b_o^2 - b_i^2 = \frac{F_o + F_o S \cdot 12 d_o^2}{\pi^2 E (b_o^2 + b_i^2)}$$

$$m = \rho A_c d_o$$

$$= \rho \frac{F_o + F_o S \cdot 12 d_o^2}{\pi^2 E (b_o^2 + b_i^2)} d_o = \frac{12}{\pi^2} F_o + F_o S \cdot \frac{d_o^3}{b_o^2 + b_i^2} \cdot \frac{\rho}{E}$$

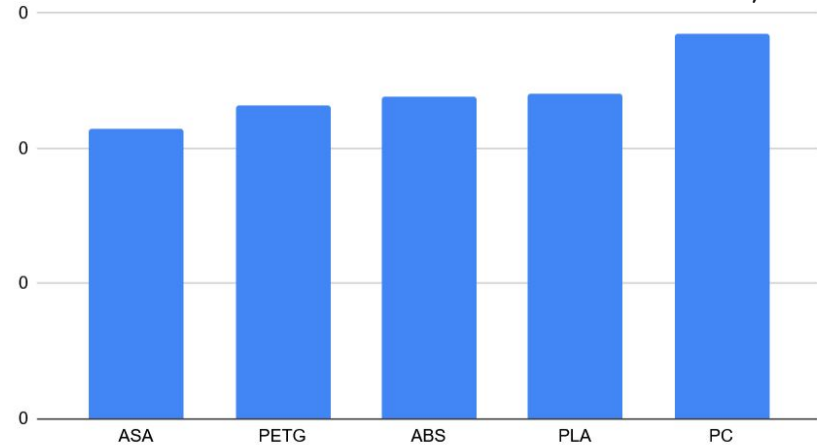
$$M = \frac{E}{\rho}$$

↑

$$\frac{\rho}{E}$$

- Multiple material factors (compression, bending, buckling)
- Legs are heaviest part so buckling factor is critical

Density / Modulus for Thermoplastics (3D printed material data from Bambu)



ASA minimizes this factor out of accessible thermoplastics

Leg Failure and Mass Analysis

buckling ← main failure mode (hollow square legs)

$$F_{cr} = \frac{C \pi^2 E I}{l^2}$$

$$F_{at} F_{oS} = \frac{\pi^2 E (b_o^4 - b_i^4)}{12 d_o^2}$$

$$b_o^4 - b_i^4 = \frac{F_{at} (F_{oS}) (12) (d_o^2)}{\pi^2 E}$$

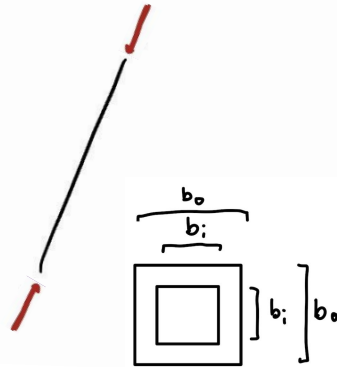
$$b_o^2 - b_i^2 = \frac{F_{at} F_{oS} 12 d_o^2}{\pi^2 E (b_o^2 + b_i^2)}$$

$$m = \rho A_c d_o$$

$$= \rho \frac{F_{at} F_{oS} 12 d_o^2}{\pi^2 E (b_o^2 + b_i^2)} d_o = \frac{12}{\pi^2} F_{at} F_{oS} \cdot \frac{d_o^3}{b_o^2 + b_i^2} \cdot \frac{\rho}{E}$$

$$I = \frac{b_o^4 - b_i^4}{12}$$

Assume pin-pin;
 $e = 1$



$$M = \frac{E}{\rho}$$

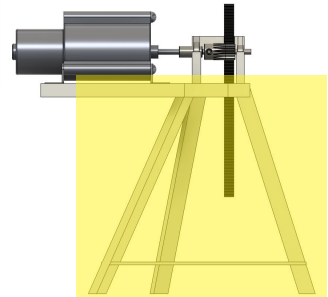
↑

↳ compression failure mode:

$$\sigma = \frac{F}{A}$$

$$\frac{S_y}{F_{oS}} = \frac{F_{at}}{b_o^2 - b_i^2}$$

$$b_o^2 - b_i^2 = \frac{F_{at} (F_{oS})}{S_y}$$



Platform Failure Analysis

$$\sigma = \frac{My}{I}$$

$$\frac{S_y}{FOS} = \frac{My}{I}$$

$$\frac{S_y}{FOS} = \frac{P_e(l_s)(\frac{t}{2})}{\frac{1}{12}wt^3}$$

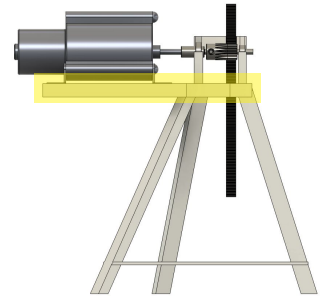
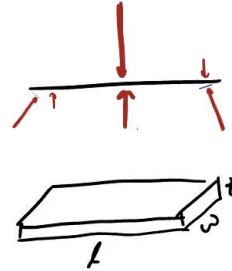
$$S_y = \frac{6FOS F_e l_s t}{wt^3}$$

$$wt^3 = \frac{6FOS F_e l_s}{S_y}$$

$$m = \rho lwt$$

$$= \rho l \frac{6FOS F_e l_s}{S_y t} = 6FOS F_e \cdot \frac{l l_s}{t} \cdot \frac{\rho}{S_y}$$

$$M = \frac{S_y}{\rho}$$



Web Failure and Mass Analysis

→ axial tension failure

$$\sigma = \frac{F}{A}$$

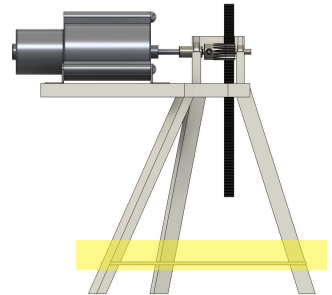
$$\frac{\delta_y}{FoS} = \frac{F_w}{A}$$

$$A = \frac{F_w FoS}{\delta_y}$$

$$M = \frac{S_y}{\rho}$$

$$m = \rho A l$$

$$= \rho \frac{F_w FoS}{\delta_y} l = F_w FoS \cdot l \cdot \frac{\rho}{S_y}$$



Housing Inverse Analysis

$$\frac{179.5}{0.0086 \cdot 0.0041}$$

Bearing 1 Compression

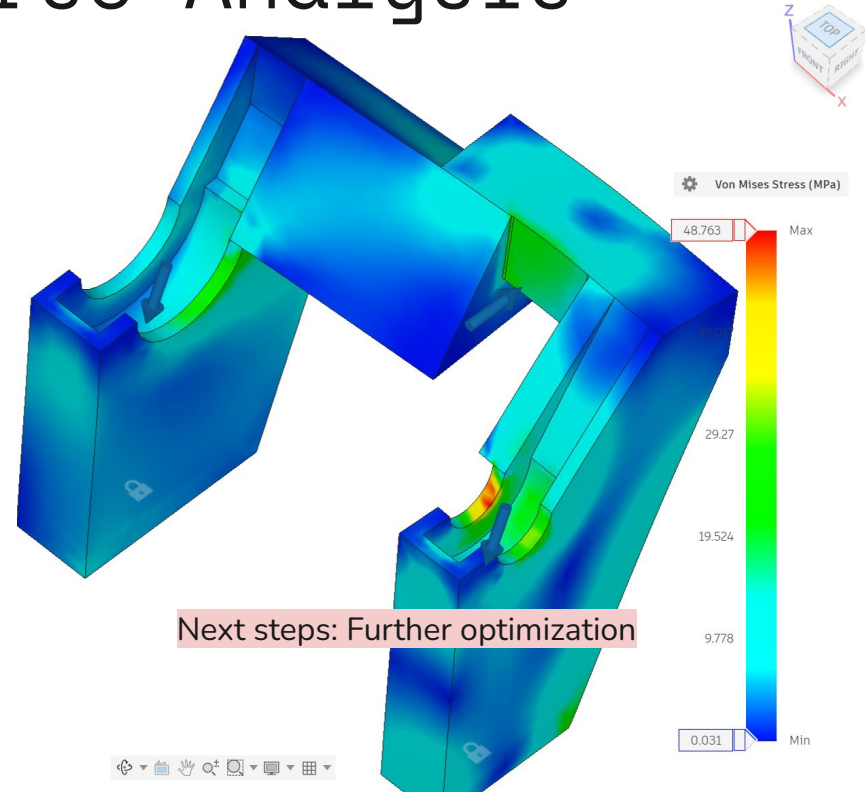
$$= 5090754.39592$$

$$\frac{162.5}{0.0046 \cdot 0.0116}$$

Bearing 2 Compression

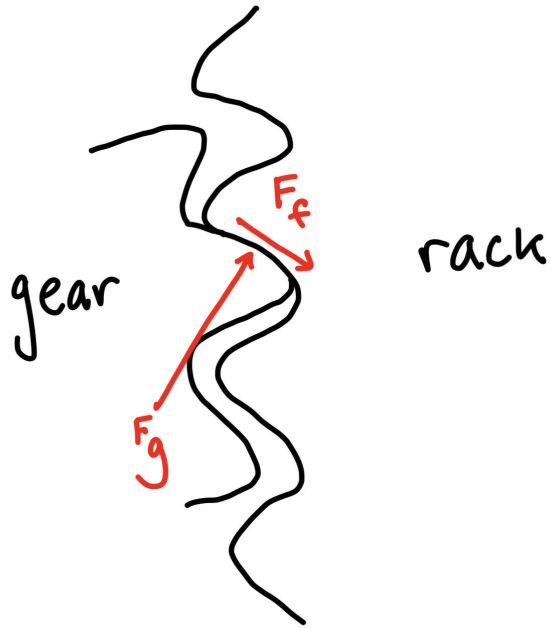
$$= 3045352.32384$$

ASA Yield: 35 MPa



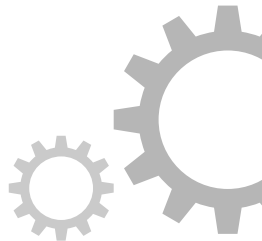
Transmission Energy Loss

This term is 0 for the rack!



$$\eta \approx \frac{1 - \mu \cdot \left(\alpha + \frac{\pi}{N_2} \right)}{1 - \mu \cdot \left(\alpha - \frac{\pi}{N_1} \right)}$$

$\mu = 0.1 \rightarrow 97.8\%$ efficiency



Rack Energy Loss

Racks

$$F_f = \mu N_{\text{rack}} = \mu F_{\text{gear}} \sin \theta_{\text{pressure}}$$

$$F_f = 0.10 (165) \sin 20^\circ = 5.07 \text{ N}$$

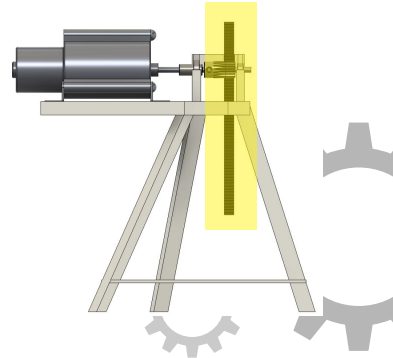
ASA-Steel, lubricated

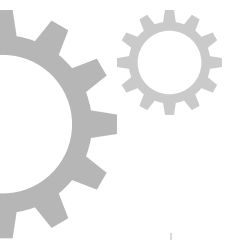
$$\mu = 0.10$$

He et. al, Polymers, 2023

$$5.07 \text{ N} * 0.05 \text{ cm} = 0.25 \text{ J}$$

12 N at stall $\rightarrow$ 3.5% impulse decrease





Bearing Energy Loss

Bearing

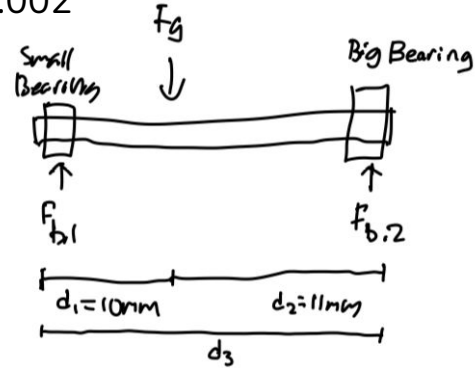
$$\sum M = d_3 F_{b,2} - d_2 F_g = 0$$

$$F_{b,2} = \frac{d_1 F_g}{d_3} = \frac{(365)(10 \times 10^{-3})}{(10 + 11) \times 10^{-3}} = 173.81 \text{ N}$$

$$\sum M = d_3 (-F_{b,1}) + d_2 F_g = 0$$

$$F_{b,1} = \frac{d_2 F_g}{d_3} = \frac{(365)(11 \times 10^{-3})}{(10 + 11) \times 10^{-3}} = 191.14 \text{ N}$$

$$\mu = 0.002$$



$$(174 \cdot 0.003 \cdot 0.002) \cdot 12.5$$

Torque (B2)

Distance

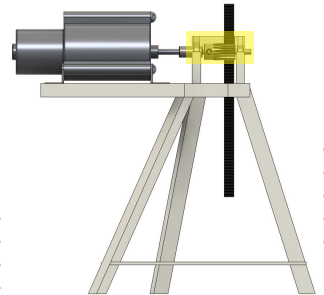
$$= 0.01305$$

$$(191 \cdot 0.006 \cdot 0.002) \cdot 12.5$$

Torque (B1)

Distance

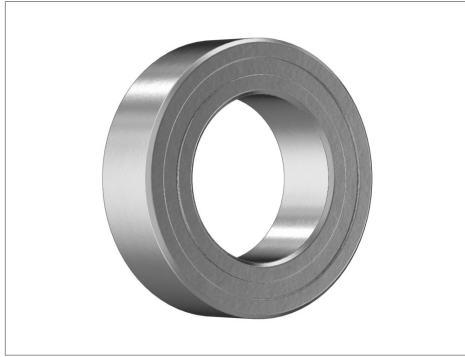
$$= 0.02865$$



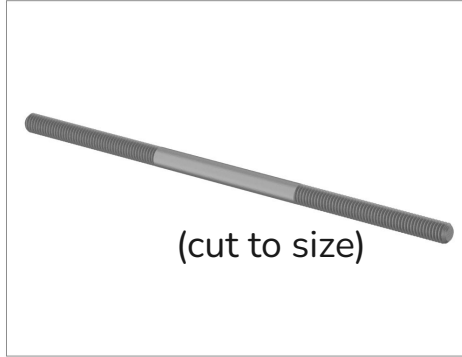


Shaft & Bearing Selection

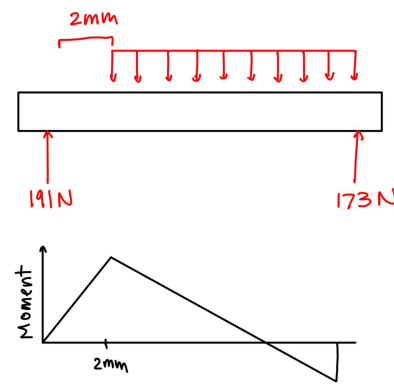
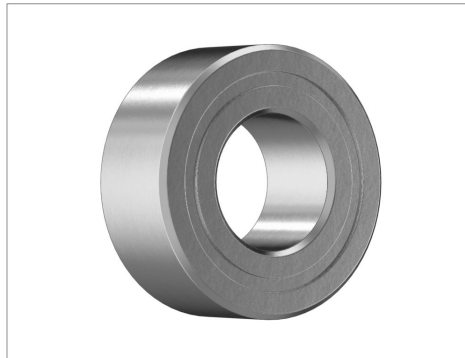
Corrosion-Resistant 440C Stainless Steel Ball Bearing
Shielded, Trade No. 106-2Z, for 6 mm Shaft Diameter



Connecting Rod
Carbon Steel, 75 mm Overall Length, M3 x 0.5 mm Thread



Corrosion-Resistant 440C Stainless Steel Ball Bearing
Shielded, Trade No. 63-2Z, for 3 mm Shaft Diameter



Shaft Bending FOS:

$$\frac{\left((191 \cdot 0.004) \cdot \frac{0.003}{2} \right)}{\pi \cdot \frac{\left(\frac{0.003}{2} \right)^4}{4}} - 1 \cdot 500 \cdot 10^6$$

= 1.73476361353

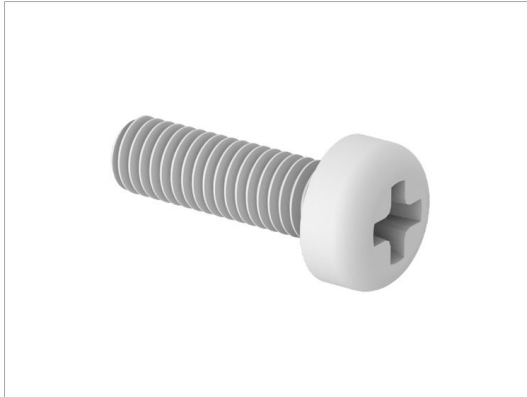


Fastener Selection

Axial Load $\frac{34}{\left(\frac{0.003}{2}\right)^2 \cdot \pi} = 4810016.05789$

Nylon Yield: 70 MPa

Nylon Pan Head Phillips Screws
M3 x 0.50 mm Thread, 10mm

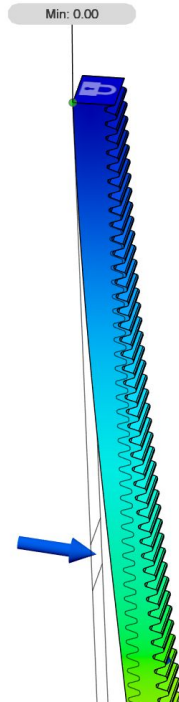


7mm platform → 5 free threads

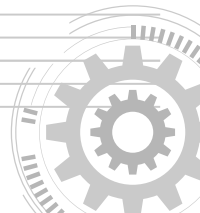
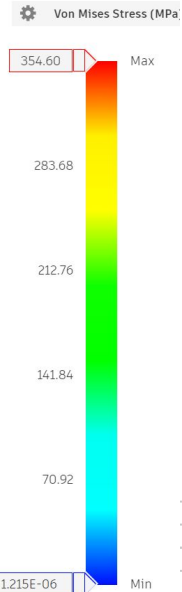
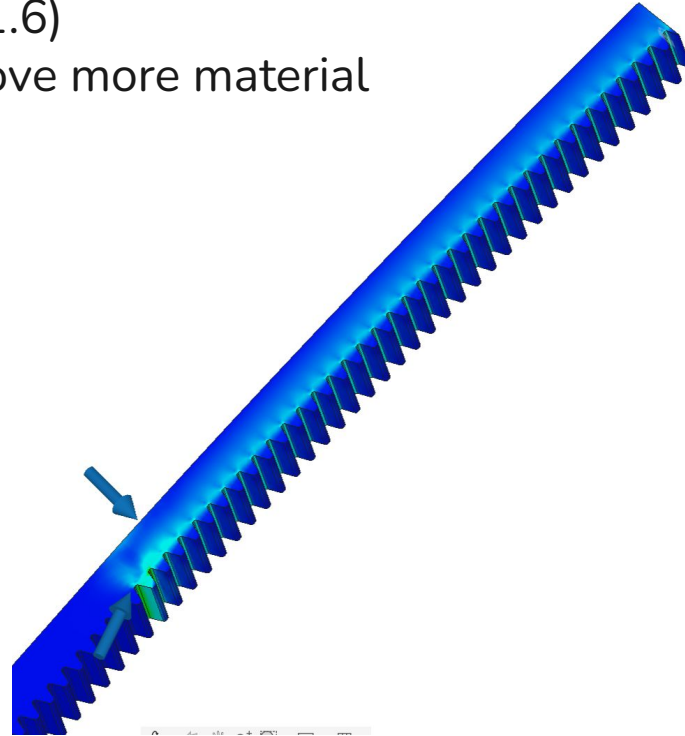
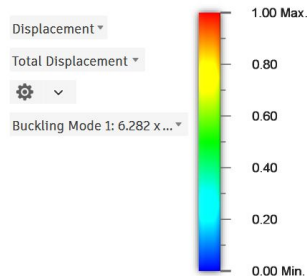
Rack Loading FEA

High FOS except on teeth (~1.6)

6.2 buckling FOS → can remove more material

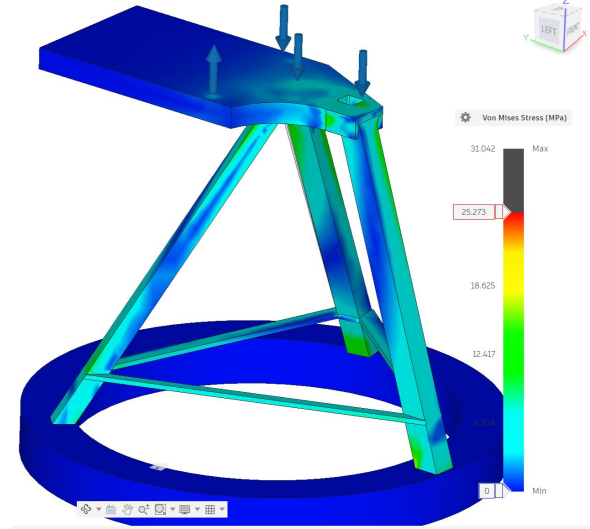


Fixed-free



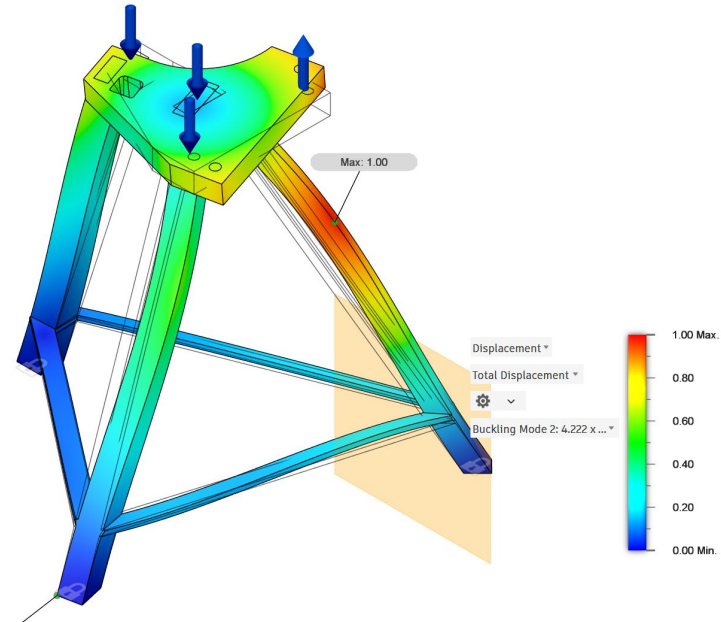
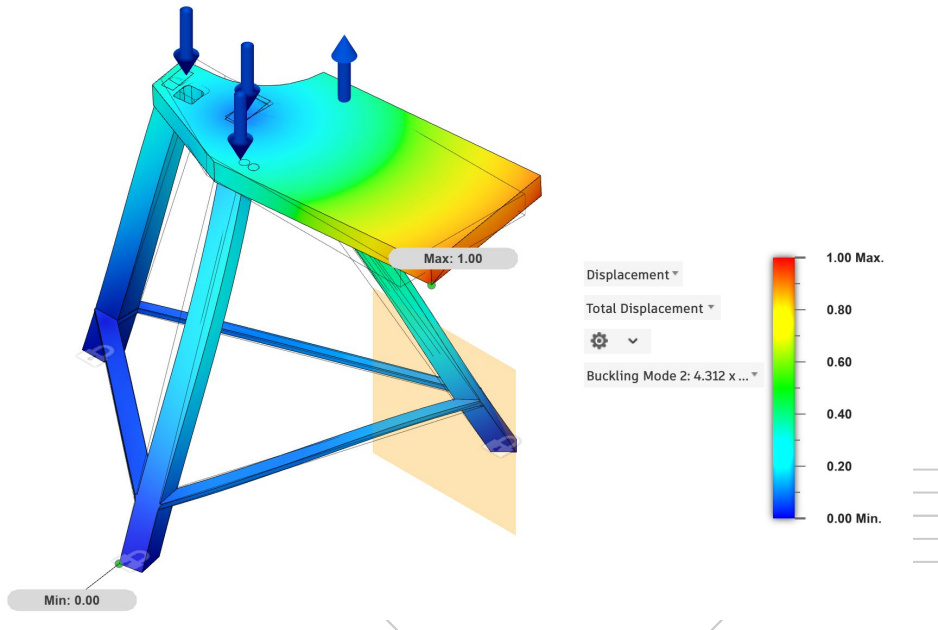
Structure Static Stress

- Some combined loading in legs and web
- Even stress in a, b, c
- 2 FOS in most places, with some concentrations to optimize



Structure Buckling

- With platform, mode is physically unreasonable
- Cut-down platform is as expected (4 FOS)



Conclusion

- Design:
 - Rack & Pinion
 - Asymmetrical Tripod
 - Tensile Web support
- $FS_{\text{stock parts}} > 1.5$
- $FS_{\text{custom parts}} = 2$
- Mass: 61 g (w/o motor & gearbox)
- Impulse: 2015 Ns

